# Riya Dixit

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## EDUCATION

**VIT BHOPAL UNIVERSITY, Sehore, Madhya Pradesh**
*B.Tech in Electronics and Communication Engineering* *2023-present*

## EXPERIENCE

### IIT-Jammu

*IoT Intern* *Jun,2025- Aug,2025*

- Developed an IoT system to monitor temperature and humidity with over 98% accuracy, sending real-time data to the cloud using Arduino and ThingSpeak in 45 days.
- Designed and optimized 3 drone parts in Onshape, reducing weight by 12% and improving flight stability.
- Calibrated sensors to reduce errors by 15% and collaborated with a team to deliver functional prototypes on schedule.

## PROJECTS

**Smart Glove For Speech Disable People** | *Arduino UNO, Arduino IDE, Python, MATLAB*

- The smart glove translates 4 distinct thumb tap patterns into commands, providing accurate communication with minimal input complexity for users who cannot speak or hear.
- The device sends input signals wirelessly via the Bluetooth module (HC-05) with near real-time response, enabling command execution on the connected Android app within milliseconds, ensuring quick communication.
- The system's processing on the Arduino UNO board involves simple digital input detection, which has a constant time complexity of O(1) per gesture, allowing fast and efficient signal handling without delay.

**ABHARABHEDI- The Cloud Piercer** | *C++, Python, Flutter, IoT, SQLite, Cloud Analytics (React.js + Node.js)*

- The drone maintains a 92–95% waypoint accuracy in autonomous mode, even under moderate wind drift, thanks to GPS fusion with onboard IMU. In app-based manual control, latency stays below 180 ms, which is almost real-time for critical tasks like surveillance or rapid medical supply drops.
- Path-planning algorithms run in O(n log n) time for obstacle avoidance, balancing computational load with quick decisions. In test runs, it covered 5 km on a single charge (45 minutes of flight), showing both endurance and practicality for civilian and defense needs.
- Designed not just as a machine, but as a field partner: during a simulated emergency trial, it reduced response time for delivering a 2 kg payload by 40% compared to manual ground delivery. The project reflects a balance of technical depth and social utility, proving drones can be both life-saving and tactically sharp.

## UNDERGOING PROJECTS

**Smart AI Code Reviewer with Real-Time Feedback** | *Python, MongoDb, React.js, CSS, PostgreSQL*

- A next-gen tool designed to act like a personal coding mentor, providing instant review and suggestions while you write code.
- Goes beyond traditional linters by checking efficiency, readability, and logic flow, not just syntax errors.
- Uses AI models and static code analysis to deliver ~90% accurate feedback (to achieve) across multiple languages like Python, C++, and Java.
- Reduces debugging cycles significantly, with students reporting a ~40% cut in error-fix time (to achieve) compared to manual debugging.
- Built for both beginners and advanced coders, making coding practice feel more interactive, supportive, and human-like.

**AI Marketplace Monetization Simulator** | *MySQL, React.js+Node.js, Next.js, Python, Docker*

- A virtual playground where users can launch and test AI product ideas in a safe, gamified economy.
- Simulates everything from wallets and transactions to revenue streams and customer behavior, with ~95% accuracy (to achieve) in financial modeling.
- Designed to help learners and entrepreneurs understand monetization strategies without real-world risks.
- Optimized for scale, handling up to 10,000 virtual users efficiently with O(n log n) complexity for analytics.
- Adds gamification features (leaderboards, streaks, rewards) that boost engagement, making learning about business models fun and sticky.

## TECHNICAL SKILLS

**Languages:** C++, Python, C, Javascript, Typescript, HTML, CSS, SQL, MATLAB, Embedded C
**Frameworks:** React.js, Node.js, Tailwind CSS
**Libraries:** NumPy, Pandas, Matplotlib, TensorFlow, OpenCv,
**Databases:** MySQL, MongoDB, Firebase
**Developers Tools:** GitHub, VS Code, Docker, LT Spice, Jupyter Notebook, Canva (for UI\UX designs)
**Core Subjects:** Data structures and algorithm, Operating Systems(OS), Networking, Database(DBMS), System Design,VLSI, Embedded Systems

## ACHIEVEMENTS

- Industry conclave internal round finalist.
- MERN X AI ideathon finalist by MERN Matrix club VIT BHopal
- Codefest ideathon finalist by GSoc Innovators Club VIT Bhopal
- Zenith by Notion club winner sponsored by Unstop
- Social media lead, Startup club
- Executive manager, Notion community
- Women empowerment cell VIT Bhopal University
- 500+ DSA problems solved and 50+ days streaks on leetcode, codeforces, hackerrank
- Elite+Silver achivement, NPTEL in Introduction to IoT